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MILITARY SPECIFICATION

SEMITRAILERS, REAR DUMP; SINGLE AXLE AND TANDEM AXLE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single axle and tandem axle rear dump semitrailers.

1.2 Classification. Semitrailers are of the following types, as specified (see 6.2 and 3.5.5):

Type I - Single axle, 12-ton payload capacity, loose material.
Type II - Tandem axle, 20-ton payload capacity, sand and gravel.
Type III - Tandem axle, 34-cubic yard capacity, demolition.
Type IV - Tandem axle, 30-cubic yard, U-shaped dump body.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-H-46170 - Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.
MIL-V-62038 - Vehicle, Wheeled, Preparation for Shipment and Storage of.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Automotive Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking & Data Plate Standards.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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ASTM

ASTM E380 - Metric Practice.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J318 - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers Truck, Truck-Tractors, and Trailers.
SAE J560 - Seven-Conductor Electrical Connector for Truck-Tractor Jumper Cable.
SAE J588 - Turn Signal Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width.
SAE J682 - Rear Wheel Splash and Stone Throw Protection.
SAE J700 - Fifth Wheel Kingpin-Commercial Trailers and Semitrailers.
SAE J702 - Brake and Electrical Connection Locations- Truck-Tractor and Truck-Trailers.
SAE J1292 - Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Avenue, Coply, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The semitrailer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the semitrailer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First production vehicle. The contractor shall furnish a semitrailer for first production vehicle inspection.

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3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 General design.

3.5.1 Federal Motor Vehicle Safety Standards. The semitrailer shall comply with DoT Federal Motor Vehicle Safety Standards in effect at time of manufacture.

3.5.2 Curb weight. The curb weight shall include the weight of the semitrailer in operating condition with all attachments, equipment, and accessories.

3.5.3 Gross weight. The rated gross weight shall include the curb weight and the specified payload required in table I, evenly distributed over the dump body area. Established component and vehicular ratings and capacities shall not be increased to meet the requirements of this specification.

3.5.4 Measurement systems. Unless otherwise specified, (see 6.2) either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM E380.

3.5.5 Dimensions. Semitrailer dimensions (see 3.5.4 and 6.2) shall be as specified in table I, when the floor of the semitrailer body is in travel position on level ground, without payload.

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TABLE I. Weights and dimensions.

	TYPE I	TYPE II	TYPE III	TYPE IV
Payload (lb, min)	24,000	40,000	50,000	
Cubic yard capacity (min)	-	-	34	
Overall length of semitrailer (inches (in), maximum (max))	408	446	372	360
Overall width of semitrailer (in, max)	96	96	96	96
Height of body sides, interior (in, +1, -0)	52	24	60	48
Body width, interior (min)	88	88	84	80
Body length, interior (in, +18, -0)	288	336	336	350
Ground clearance (in, min)	9.5	9.5	9.5	9.5
Ground clearance under landing gear assembly in retracted position (in, min)	14	14	12	12
Landing wheel clearance, 6 inches below centerline of the kingpin, measured rearward, without any interference (in, min)	82	82	82	82
Kingpin location, measured rearward from front of the semitrailer (in, ±6)	30	30	30	30
Height of upper fifth wheel plate (in, ±1)	51	51	51	52
Swing radius from centerline of kingpin to nearest interference point on the semitrailer (in, min)	60	60	60	60

3.6 Performance. The semitrailer, fully equipped and loaded with specified minimum (min) payload in pounds (lb) (see table I), shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unimproved roads (see 6.4.2), and over reasonably hard uneven terrain; and being towed at speeds as great as 50 mph over improved roads (see 6.4.1).

3.6.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree (°) angle to the coupled towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

3.6.2 Tracking ability. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(a).

3.6.3 Brake performance. The semitrailer service brake system shall stop the semitrailer-tractor combination with the semitrailer loaded with specified payload within the stopping distance requirements of DoT Federal Motor Carrier Safety Regulations, section 393.52.

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3.7 Chassis components.

3.7.1 Frame. The semitrailer frame shall be full length and the manufacturer's standard.

3.7.2 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, when the semitrailer is loaded with its rated payload. Clearance shall preclude interference between tires and any other part of the semitrailer under the operating conditions specified herein.

3.7.3 Axle(s). Type I semitrailer shall be furnished with a single axle. Types II, III, and IV semitrailer shall be furnished with tandem axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with semitrailer loaded with rated payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubcaps shall have a window for visual determination of oil level. Provision for venting, or other method of withstanding internal pressure buildup, and for replenishing the oil supply shall be incorporated.

* 3.7.4 Wheels and tires.

* 3.7.4.1 Wheels and tires. Semitrailer shall be equipped with dual disk type wheels on each axle. Tires shall be steel belted radial tubeless type with highway tread. The wheel and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tire and wheel sizes shall be the same for all wheels of the semitrailer. Tires shall be rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with semitrailer loaded with rated payload.

3.7.4.2 Tire carrier. One tire carrier shall be provided and installed in a readily accessible location under the chassis. Means shall be provided for securing a spare wheel and tire assembly within the carrier to prevent accidental loss.

3.7.4.3 Spare wheel. When specified (see 6.2), a spare wheel shall be mounted on the tire carrier.

3.7.4.4 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel specified in 3.7.4.3. The spare tire shall be of the same size and tread design and of the same ply rating as tires installed on the semitrailer.

3.7.5 Brakes. Brakes shall be of the full air operated, internal expanding type and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40, 393.43, and 393.45 through 393.47, and as specified herein. The braking system shall include manual or automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and all other components required for a complete airbrake system. Gladhands shall conform to SAE J318. Air hose location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and

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protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

3.7.5.1 Parking brakes. The semitrailer shall be provided with coil spring actuated type parking brakes. The parking brakes shall be automatically applied upon disconnection to the emergency air line and under emergency braking conditions. The parking brakes shall hold the semitrailer, with rated payload, on a 20 percent grade despite the depletion of the compressed air supply. Parking brakes shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.41.

3.7.6 Upper fifth wheel plate. The upper fifth wheel plate shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.70. The kingpin shall be of heat treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate shall have turned-up lip for ease of coupling.

* 3.7.7 Lighting. The electrical lighting system shall be 12-volt (V) direct current (dc) design. The lighting system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32, and 393.33. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails, unless recessed and fully protected, or semitrailer bumpers. The front of the semitrailer shall be equipped with a receptacle conforming to SAE J560, with the receptacle located and the conductors connected and color coded as specified herein. All electrical wiring shall conform to SAE J1292. Turn signal lamps shall conform to SAE J588.

3.7.8 Landing gear. Semitrailer shall have two vertical lift, telescopic, nonrotating landing legs, with two speed gears and a handcrank on the curbside. Landing legs shall be equipped with steel wheels or self-leveling skid pads. Supports for the crank extension shafts and clips for holding cranks when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. The landing gear shall have a range of adjustment to vary the height of the upper fifth wheel plate, at centerline of kingpin, from 45 inches to not less than 53 inches from the ground. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall be not less than specified in table I.

3.7.9 Support jacks. When specified (see 6.2), two hinged type support jacks shall be provided for type I and type II semitrailers at the front of the semitrailer. When in static position, the jacks shall be capable of supporting their share of the load imposed, when semitrailer is loaded with rated payload.

3.7.10 Rear wheel splash and stone throw protection. Rear wheels shall have mud flaps at rear. Splash and stone throw protection shall be in accordance with SAE J682. When specified (see 6.2), steel fenders that cover the top and a minimum 15° radius of the forward portion of the tires shall be furnished.

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3.8 Body components.

3.8.1 Dump body. The semitrailer shall be equipped with a rear dump body from which the load is gravity dumped by opening the rear tailgate and raising the front end of the body. The body and the tailgate shall be of all welded fabrication, utilizing high-tensile steel having a minimum yield point of 45,000 pound-force per square inch for all plate and reinforcing structural members. Sides shall be of plate that is corrugated, bowed, or reinforced by wrap around or vertical braces of the pyramid, triangular or box types, to withstand their proportion of the body and of the maximum payload of the semitrailer. The sides of type II semitrailer shall have standard 2-3/4-inch slots for adding wooden or steel sideboards. The floor shall be of butt-welded metal plates reinforced by crossmembers or interlaced under-structure, to withstand its proportion of the maximum payload of the semitrailer. The tailgate and its hinge components shall withstand the force of the specified payload when the body is raised to its maximum dump angle with the tailgate closed. The height for both the front end and the tailgate of type II semitrailer shall exceed the height of the sides. The body front and sides of type III semitrailer shall be constructed of not less than 0.1345-inch (U. S. revised standard gage No. 10) thick steel, and the floor and tailgate of not less than 0.1793-inch (U. S. revised standard gage No. 7) thick steel. The interior of the body shall have no obstructions which will impede the flow of the material during dumping. A front box or hoist housing is acceptable. The dump shall be U-shaped for type IV semitrailer and be the manufacturers standard commercial product.

3.8.1.1 Tailgate, types I, II, and IV . The type I, type II and type IV semitrailer shall be provided with a double-acting tailgate, hinged at both top and bottom. The tailgate shall be furnished with chains to permit the tailgate, hinging at the top, to open completely when dumping; to permit the tailgate, hinging at the top, to have a choice of openings at the bottom for spreading of aggregate; and to allow the tailgate, hinging at the bottom, to be laid flat with the floor of the body. Unless otherwise specified (see 6.2), the tailgate shall be opened at the bottom by a driver- actuated, tailgate-trip control installed on the left front of the body. Release of the tailgate for dumping a heaped load shall not require driver effort in excess of 60 pounds at the control. Air controls, when furnished, shall be protected from falling aggregate.

3.8.1.2 Tailgate, type III. The type III tailgate shall be of the side-hinged, barn door type with outside latches. The tailgate shall be reinforced with vertical and horizontal high strength structural members.

3.8.1.3 Tarpaulin. When specified (see 6.2), a tarpaulin of sufficient size to cover the top of the body with overlap on the sides shall be furnished. The tarpaulin shall be of 16 ounce neoprene rubber coated nylon cloth with grommets or tie loops on 24-inch centers. Tarpaulin tie-down rails shall be provided. The required number of rubber tie-down straps with hooks at both ends shall be provided to secure the tarpaulin over the body.

3.8.1.4 Tarpaulin rack. When tarpaulin is specified (see 3.8.1.3), a tarpaulin storage rack shall be installed on the front of the body. The rack shall be a shelf or basket type and shall be fabricated of metal. A means of storage for the required number of tie-down straps shall be provided.

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* 3.8.2 Hydraulic system. The semitrailer shall be equipped with an engine powered source (see 3.8.4), hydraulic body hoist system. The hydraulic system shall be complete with all operating accessories, including pump(s), reservoir, hydraulic cylinders, valves, pressure gages, pressure relief valves, filter, hoses, piping, and fittings. The dump body shall hoist to the dump angle of the manufacturer's standard semitrailer. Hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system. A permanent decal near the filler opening shall be installed with this statement: "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-46170 ONLY". At least 6- 1/2 inches of clearance shall be provided between exhaust systems. The cylinder rams or telescopic tubes shall be ground and polished.

3.8.2.1 Hoist system. The hoist system shall consist of components rated to perform the hydraulically powered operations of the dump body in accordance with the specified payload. Mechanisms and reservoirs shall be positioned or guarded to prevent any possible hazard or damage from hot surfaces, debris, or obstructions. The dump hoist control shall be the three-way type, providing hoist, lower, and hold.

3.8.2.2 Hoist actuation. The actuation of the body hoist system shall be the self-contained type, whereby the complete hoist system shall be mounted on the semitrailer with an auxiliary engine driving hydraulic pump. The engine and hydraulic pump components shall be mounted on the gooseneck of the semitrailer.

3.8.2.3 Hoist system controls. The hoist system controls shall be mounted on the street side front of the semitrailer and adjacent to the engine controls. The controls shall be located to permit operation by the operation from a standing position on the ground.

3.8.3 Shroud. Engine shall be provided with a sheet metal shroud, with side panels, to protect the engine when loading the semitrailer and during inclement weather conditions. Panels shall be removable. Louvers shall be provided for air circulation.

* 3.8.4 Power source. A standard commercial, four-stroke cycle, air-cooled gasoline or diesel engine shall be provided to drive the hydraulic pump assembly. The engine shall be mounted in conjunction with the hydraulic pump. Mounting shall be so as to minimize transfer of vibration. The engine shall have the horsepower, torque, and speed characteristics to meet satisfactorily all performance requirements specified herein.

* 3.8.4.1 Engine components. The engine shall be equipped with as a minimum the following components:

a. Ignition system.
b. Electric starting system, 12V dc.
c. Manufacturer's standard air cleaner.
e. Manual choke (gasoline).
f. Battery, 12V dc.
g. Governor.
h. Manually operated hand throttle.
i. Ammeter or charging indicator.
j. Fuel gage or indicator.

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* 3.8.4.2 Engine control panel. An engine control panel shall be provided. The key-operated ignition switch, hand throttle, and ammeter or charging indicator shall be installed in the control panel.

3.8.4.3 Engine fuel tank. An engine fuel tank shall be furnished. The fuel tank cap shall be attached to the tank by means of a corrosion-resistant chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. A shutoff valve shall be provided in the line to the power source and a decal shall be furnished and marked denoting the type of fuel to be used, and shall be located in a conspicuous location near the filler cap.

3.8.4.4 Exhaust system. The exhaust system shall be furnished with a muffler and a rain cap when pipe ending is upright and means shall be provided to vent exhaust fumes away from the operating personnel. The exhaust system shall be shielded to prevent contact of hydraulic oil in the event of a line rupture or leaking in the hydraulic system.

3.9 Treatment, painting, and data plates. Treatment, painting, and data plates shall be in accordance with MIL-STD-1223, as specified for the Navy, except the exterior color shall be gloss yellow No. 13538 of FED-STD-595.

3.10 Marking. Marking and registration numbers shall be in accordance with MIL-STD-1223, as specified for the Navy. The gross weight shall be marked on each forward front side of the body.

* 3.11 Rustproofing. When specified (see 6.2), the semitrailer shall be rustproofed in accordance with FED-STD-297.

* 3.12 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government, the contractor shall service and adjust the semitrailer for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes (when required).
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement (when required).
e. Adjustment of engine to include tune-up (when required).
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.

The semitrailer shall be conspicuously tagged to identify the lubricants and their temperature range.

3.13 Workmanship.

3.13.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.13.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the tests of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

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4.3 Examination. Each semitrailer shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle tests.

4.4.1 Vehicle weight. Semitrailer shall be weighted to determine net weight and distribution of net weight on fifth wheel and axle(s). The imposed loading on the fifth wheel and axle(s) shall be computed using the net weight and the payload. Calculated imposed loads on fifth wheel and axle(s) shall be utilized to ascertain that the suspension, axle(s), and tires furnished are of adequate capacity to meet requirements specified in 3.7.2, 3.7.3, and 3.7.4.1.

4.4.2 Road test. Road test shall consist of coupling the semitrailer loaded with rated payload to a truck-tractor. The tractor combination shall be driven a distance of not less than 20 miles under the conditions specified in 3.6. At least 20 percent of the distance shall be over hard uneven terrain. At least five sudden stops shall be made from a speed of not less that 20 mph. Turning ability shall be observed to determine conformance to 3.6.1, and tracking ability shall be observed to determine conformance to 3.6.2. After the test, the semitrailer shall be examined for evidence of misalinement, binding, or other malfunction.

4.4.3 Landing gear test. The landing gear shall be tested to verify conformance to 3.7.8.

4.4.4 Support jacks test. The semitrailer, when furnished with support jacks, shall be tested to verify conformance to 3.7.9.

4.4.5 Dump test. The semitrailer shall be cycle tested three times, load and dump. Semitrailer shall show no malfunction, permanent deformation, leakage, or binding of the dumping mechanism.

4.4.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.5 Production semitrailer tests. The contractor's testing system shall, as a minimum, assure that the semitrailer is capable of meeting the performance requirements specified herein.

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4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

* 5.1 Vehicle processing. The equipment shall be preserved and packed in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packed, and marked in accordance with the requirements of MIL-V-62038 with the level of preservation and packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The semitrailers are intended to transport various types of materials on and off site on constructions jobs.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of semitrailer required (see 1.2 and 3.5.5).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When only one system of measurement (US or SI) is acceptable (see 3.5.4).
e. When spare wheel is required (see 3.7.4.3).
f. When spare tire is required (see 3.7.4.4).
g. When support jacks are required (see 3.7.9).
h. When rear wheel fenders required (see 3.7.10).
i. When the tailgate operation is to be different from that specified (see 3.8.1.1).
j. When tarpaulin is required (see 3.8.1.3).
k. When rustproofing is required (see 3.11).
l. When preservation-packing in accordance with MIL-V-62038 is required and the level of preservation and packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 Definitions.

6.4.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

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6.4.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Subject term (key word) listing.

Disposal
Payloads
Productivity

Preparing Activity:
Navy - YD

(Project 2330-N100)

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
A. VILLEGAS	(805) 982-5773	551-5773
c. ADDRESS (Include Zip Code) CIVIL ENGINEERING SUPPORT OFFICE NAVAL CONSTRUCTION BATTALION CENTER PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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